

Reservoir Architecture

Bigram encoder and double-delay reservoir

This model represents a simple recurrent reservoir fed by a bigram encoder. A readout node (predictive hetero-associative memory component) learns to predict the next input based on the current input and the state of the reservoir.

This architecture functions as a basic sequence memory.

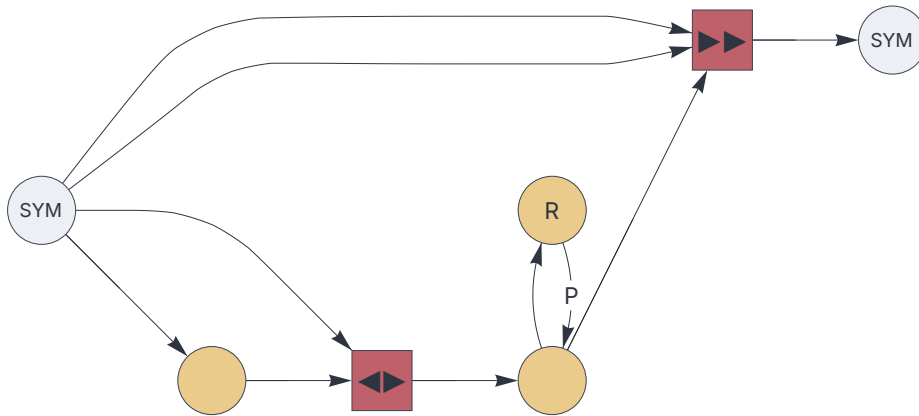
Requirements

```
<< "Circuits.m"
```

Circuit

```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",
  "hyperparameters": {"default": [2000, 8]},
  "dataflow": [
    {"component": "input", "plugin": "codec", "send": [11]},
    {"component": "delay", "send": [21], "receive": [11]},
    {"component": "heteroencoder", "send": [51], "receive": [11, 21]},
    {"component": "delay", "send": [61], "receive": [[51, {"slot": 62, "tags": ["permute"]}]]},
    {"component": "delay", "send": [62], "receive": [61], "rate_limit": 3.0},
    {"component": "predictor", "send": [101], "receive": [11, [11, 61]]},
    {"component": "output", "plugin": "codec", "receive": [101]}
  ]
}
```

```
circuit = Circuit[json];
c = circuit["function"];
circuit["schematics"][ImageSize->500]
```



Sequence learning

```
{Null, Null, p, Null, e, Null, p, {l, p}, e, A,
 p, p, l, e, A, p, p, l, e, A, p, p, l, e, A, p, p, l, e, A}
```

Dataset

A sequence of 1000 digits:

```
dataset = RandomInteger[ {1, 10}, 1000];
```

Prediction score

```
Do[
  predictions = c /@ dataset;
  Print["  iteration = ", i,
    ", prediction score = ", N[Mean@Boole@MapThread[SameQ,
      {dataset, RotateRight[predictions]}]]],
    {i, 1, 5}];
```

```
iteration = 1, prediction score = 0.101  
iteration = 2, prediction score = 0.995  
iteration = 3, prediction score = 1.  
iteration = 4, prediction score = 1.  
iteration = 5, prediction score = 1.
```